

# MITHILESH ADHINARAYANAN

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## PROFESSIONAL SUMMARY

High-achieving Computer Science (AI & Data Science) student at SASTRA University with hands-on experience in software development, Python, Java, SQL, and Machine Learning. Quantitative Researcher at WorldQuant with a proven track record of engineering profitable algorithmic trading alphas. Research based forex models (52%+ net return). Skilled in applying SDLC principles, data pipelines, and analytical problem-solving to build scalable, real-world software systems. Strong background in risk-aware system design and Agile development practices.

## EDUCATION

**SASTRA Deemed University** | B.Tech — Computer Science Engineering (AI & Data Science) *Aug 2024 – Jun 2028* Thanjavur, TN

- CGPA: **8.16 / 10.0** | Core: Data Structures & Algorithms, OOP, DBMS, Multi-Agent Systems, High-Performance Computing

**Harvard University (CS50)** | Web Programming with Python & JavaScript (CS50W) *In Progress* Remote

## PROFESSIONAL & RESEARCH EXPERIENCE

**WorldQuant** | BRAIN Research Consultant (prev. Conditional Consultant) *Apr 2026 – Present* Remote

- Engineered logic-based quantitative alphas using Custom Expression Languages; applied statistical modelling to construct, evaluate, and backtest financial signal strategies across global equities.
- Achieved **Gold Genius Level** on WorldQuant BRAIN for consistent, profitable alpha generation — demonstrating systematic analytical problem-solving and risk-aware decision-making.

**XYlofy AI** | AI & Data Science Intern *Jun 2026 – Jul 2026* Bengaluru, India

- Developed and deployed scalable machine learning architectures for real-world automation projects following Agile SDLC, collaborating with industry mentors to optimise end-to-end Python data pipelines.
- Integrated REST APIs and ML model backends; improved data throughput and reduced pipeline latency using systematic software engineering practices.

**SASTRA Deemed University** | Undergraduate Researcher *Feb 2026 – Present* Thanjavur, TN

- Quantitative Finance (IEEE Submitted):** Authored paper on USD/INR exchange rate prediction using a novel Spatial-Temporal Proximal Policy Architecture (ST-PPA); achieved 95%+ predictive accuracy and 52.94% net return (Sharpe Ratio: 0.560) — perfectly backtested with rigorous risk controls.
- Cybersecurity & Risk (APT Detection):** Leveraged Deep Learning models to analyse network traffic, dynamically identify cyber vulnerabilities, and develop risk-aware threat detection frameworks for enterprise security.

## TECHNICAL PROJECTS

**RL-Based Traffic Management System** | DQN, MARL, OpenCV, C++, Raspberry Pi, Python *Sep – Dec 2025* SIH 2025

- Architected a Multi-Agent Reinforcement Learning (MARL) control system to minimise intersection clearance times; applied full SDLC from design to deployment on constrained edge hardware.
- Optimised computationally heavy OpenCV vehicle classification algorithms in C++ and Python for sub-100ms real-time inference on Raspberry Pi.

**Web-Based AI Applications** | Python, Django, React, MongoDB, REST APIs, ML *Jan 2026 – Present*

- Engineered full-stack web platforms integrating custom ML backends; applied Agile SDLC from requirements to deployment, delivering REST APIs and interactive data visualisation dashboards.

## TECHNICAL SKILLS

**Languages:** Python (Advanced), Java, C++, C, SQL, HTML/CSS

**AI/ML & Deep Learning:** Reinforcement Learning (PPO, DQN, MARL), Neural Networks, TensorFlow, PyTorch, Transfer Learning, Time Series, Model Optimisation

**NLP & Computer Vision:** LLMs (Transformers, BERT, GPT, RAG), Prompt Engineering, Embeddings, OpenCV, YOLO, Object Detection, Image Segmentation

**Web & Systems(learning):** Django, React, Flask, MongoDB, REST APIs, SDLC, Git, Linux, HPC, JavaScript, Raspberry Pi, VS Code

**Data & Quantitative:** NumPy, Pandas, Matplotlib, Feature Engineering, EDA, Alpha generation, Financial Signal Modelling, Risk Analysis

## CERTIFICATIONS & ACHIEVEMENTS

- LFX Mentorship (Linux Foundation)** : Open Source Certification | **Yale University** : Financial Markets (Coursera) | **NPTel SWAYAM** — Deep Learning Specialisation
- Cisco** : Data Science & Networking | **Kaggle** : Data Science & ML | **Harvard CS50W** : Web Programming (In Progress)
- CMI STEMS 2025:** Selected in the **Top 30 in India** for research aptitude — Chennai Mathematical Institute.
- WorldQuant BRAIN: Gold Genius Level** — consistent profitable quantitative alpha generation across global equities.
- IOQM:** Merit Certificate holder — Indian Olympiad Qualifier in Mathematics.
- IEEE Research:** Paper submitted — DNN-RL exchange rate prediction (ST-PPA); currently under editorial review.